

Documentatie si raspunsuri UDP

Exercitiul 1:

Filtram dupa „udp” si selectam primul pachet.

Packet numberul acestui segment este 5.

Protocol message este: SSDP.

Antetul UDP are 4 fielduri: Source Port, Destination Port, Length, Checksum.

udp						
No.	Time	Source	Destination	Protocol	Length	Info
5	2.764896	10.0.0.254	239.255.255.250	SSDP	317	NOTIFY * HTTP/1.1
6	2.765750	10.0.0.254	239.255.255.250	SSDP	317	NOTIFY * HTTP/1.1
7	2.766303	10.0.0.254	239.255.255.250	SSDP	326	NOTIFY * HTTP/1.1
8	2.766974	10.0.0.254	239.255.255.250	SSDP	326	NOTIFY * HTTP/1.1
9	2.767715	10.0.0.254	239.255.255.250	SSDP	381	NOTIFY * HTTP/1.1
10	2.768361	10.0.0.254	239.255.255.250	SSDP	381	NOTIFY * HTTP/1.1
11	2.769103	10.0.0.254	239.255.255.250	SSDP	391	NOTIFY * HTTP/1.1
12	2.769651	10.0.0.254	239.255.255.250	SSDP	391	NOTIFY * HTTP/1.1
15	3.325064	10.0.0.44	75.75.75.75	DNS	77	Standard query 0x3c29 A gaia.cs.umass.edu
17	3.348972	75.75.75.75	10.0.0.44	DNS	93	Standard query response 0x3c29 A gaia.cs.umass.edu A 128.119.245.12
30	3.427392	10.0.0.44	75.75.75.75	DNS	83	Standard query 0xeda4 A maxcdn.bootstrapcdn.com
31	3.428514	10.0.0.44	75.75.75.75	DNS	79	Standard query 0xa79 A ajax.googleapis.com
35	3.445049	75.75.75.75	10.0.0.44	DNS	135	Standard query response 0xeda4 A maxcdn.bootstrapcdn.com CNAME cds.j3z9t3p6
36	3.448906	75.75.75.75	10.0.0.44	DNS	95	Standard query response 0xa79 A ajax.googleapis.com A 172.217.12.202
521	3.678228	10.0.0.44	75.75.75.75	DNS	75	Standard query 0xdcfa A www.pearson.com
522	3.678393	10.0.0.44	75.75.75.75	DNS	79	Standard query 0xb436 A www.vitalsource.com
523	3.678598	10.0.0.44	75.75.75.75	DNS	72	Standard query 0xd3a3 A redshelf.com
526	3.695928	75.75.75.75	10.0.0.44	DNS	169	Standard query response 0xdcfa A www.pearson.com CNAME wilcard.pearson.com
527	3.698647	10.0.0.44	75.75.75.75	DNS	74	Standard query 0xe1a9 A www.amazon.com
528	3.703716	75.75.75.75	10.0.0.44	DNS	159	Standard query response 0xb436 A www.vitalsource.com A 104.17.67.241 A 104.
529	3.704968	75.75.75.75	10.0.0.44	DNS	88	Standard query response 0xd3a3 A redshelf.com A 34.196.10.62
530	3.718156	75.75.75.75	10.0.0.44	DNS	169	Standard query response 0xe1a9 A www.amazon.com CNAME tp.47cf2c8c9-frontier
541	6.801907	10.0.0.44	75.75.75.75	DNS	96	Standard query 0x6cf4 A ss-prod-ue1-notif-63.aws.adobess.com
542	6.818616	75.75.75.75	10.0.0.44	DNS	144	Standard query response 0x6cf4 A ss-prod-ue1-notif-63.aws.adobess.com A 52.

Time to Live: 4			
Protocol: UDP (17)			
Header Checksum: 0xd902 [validation disabled]			
[Header checksum status: Unverified]			
Source Address: 10.0.0.254			
Destination Address: 239.255.255.250			
User Datagram Protocol, Src Port: 47931, Dst Port: 1900			
Source Port: 47931			
Destination Port: 1900			
Length: 283			
Checksum: 0xf7a8 [unverified]			
[Checksum Status: Unverified]			
[Stream index: 0]			
[Timestamps]			
UDP payload (275 bytes)			
Simple Service Discovery Protocol			
> NOTIFY * HTTP/1.1\r\n			
HOST: 239.255.255.250:1900\r\n			
0000	01 00 5e 7f ff fa 00 05	04 03 02 01 08 00 45 00	..^.....E..
0010	01 2f a1 c3 40 00 04 11	d9 02 0a 00 00 fe ef ff	./...@.....
0020	ff fa bb 3b 07 6c 01 1b	f7 a8 4e 4f 54 49 46 59	...;1...NOTIFY
0030	20 2a 20 48 54 50 2f	31 2e 31 0d 0a 48 4f 53	* HTTP/ 1.1..HOS
0040	54 3a 20 32 33 39 2e 32	35 35 2e 32 35 35 2e 32	T: 239.2 55.255.2

Exercitiul 2:

Pastram filtrul, apasam pe User Datagram Protocol si observam ca antetul are 8 bytes, apoi apasam pe fielduri si determinam ca fiecare are 2 bytes lungime.

```
▼ User Datagram Protocol, Src Port: 47931, Dst Port: 1900
  Source Port: 47931
  Destination Port: 1900
  Length: 283
  Checksum: 0xf7a8 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 0]
  > [Timestamps]
  UDP payload (275 bytes)
0010  01 2f a1 c3 40 00 04 11 d9 02 0a 00 00 fe ef ff  ./..@.....
0020  ff fa bb 3b 07 6c 01 1b f7 a8 4e 4f 54 49 46 59  ...;1...NOTIFY
0030  20 2a 20 48 54 54 50 2f 31 2e 31 0d 0a 48 4f 53  * HTTP/ 1.1..HOS
0040  54 3a 20 32 33 39 2e 32 35 35 2e 32 35 35 2e 32  T: 239.2 55.2
```

```
▼ User Datagram Protocol, Src Port: 47931, Dst Port: 1900
  Source Port: 47931
  Destination Port: 1900
  Length: 283
  Checksum: 0xf7a8 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 0]
  > [Timestamps]
  UDP payload (275 bytes)
0010  01 2f a1 c3 40 00 04 11 d9 02 0a 00 00 fe ef ff  ./..@.....
0020  ff fa bb 3b 07 6c 01 1b f7 a8 4e 4f 54 49 46 59  ...;1...NOTIFY
0030  20 2a 20 48 54 54 50 2f 31 2e 31 0d 0a 48 4f 53  * HTTP/ 1.1..HOS
0040  54 3a 20 32 33 39 2e 32 35 35 2e 32 35 35 2e 32  T: 239.2 55.2
```

```
Destination Address: 229.255.255.255
▼ User Datagram Protocol, Src Port: 47931, Dst Port: 1900
  Source Port: 47931
  Destination Port: 1900
  Length: 283
  Checksum: 0xf7a8 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 0]
  > [Timestamps]
  UDP payload (275 bytes)
0010  01 2f a1 c3 40 00 04 11 d9 02 0a 00 00 fe ef ff  ./..@.....
0020  ff fa bb 3b 07 6c 01 1b f7 a8 4e 4f 54 49 46 59  ...;1...NOTIFY
0030  20 2a 20 48 54 54 50 2f 31 2e 31 0d 0a 48 4f 53  * HTTP/ 1.1..HOS
0040  54 3a 20 32 33 39 2e 32 35 35 2e 32 35 35 2e 32  T: 239.2 55.2
0050  35 30 3a 31 39 30 30 0d 0a 43 41 43 48 45 2d 43  50:1900..
```

```
▼ User Datagram Protocol, Src Port: 47931, Dst Port: 1900
  Source Port: 47931
  Destination Port: 1900
  Length: 283
  Checksum: 0xf7a8 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 0]
  > [Timestamps]
  UDP payload (275 bytes)
0010  01 2f a1 c3 40 00 04 11 d9 02 0a 00 00 fe ef ff  ./..@.....
0020  ff fa bb 3b 07 6c 01 1b f7 a8 4e 4f 54 49 46 59  ...;1...NOTIFY
0030  20 2a 20 48 54 54 50 2f 31 2e 31 0d 0a 48 4f 53  * HTTP/ 1.1..HOS
0040  54 3a 20 32 33 39 2e 32 35 35 2e 32 35 35 2e 32  T: 239.2 55.2
```

```
Destination Address: 229.255.255.255
▼ User Datagram Protocol, Src Port: 47931, Dst Port: 1900
  Source Port: 47931
  Destination Port: 1900
  Length: 283
  Checksum: 0xf7a8 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 0]
  > [Timestamps]
  UDP payload (275 bytes)
0010  01 2f a1 c3 40 00 04 11 d9 02 0a 00 00 fe ef ff  ./..@.....
0020  ff fa bb 3b 07 6c 01 1b f7 a8 4e 4f 54 49 46 59  ...;1...NOTIFY
0030  20 2a 20 48 54 54 50 2f 31 2e 31 0d 0a 48 4f 53  * HTTP/ 1.1..HOS
0040  54 3a 20 32 33 39 2e 32 35 35 2e 32 35 35 2e 32  T: 239.2 55.2
```

Exercitiul 3:

Valoarea fieldului Length este marimea in bytes a UDP payloadului adunat cu cei 8 bytes ai antetului de UDP.

In cazul pachetului cu numarul 5:

$$\text{Length} = \text{UDP payload} + 8 = 275 + 8 = 283 \text{ bytes}$$

udp						
No.	Time	Source	Destination	Protocol	Length	Info
5	2.764896	10.0.0.254	239.255.255.250	SSDP	317	NOTIFY * HTTP/1.1
6	2.765750	10.0.0.254	239.255.255.250	SSDP	317	NOTIFY * HTTP/1.1
7	2.766303	10.0.0.254	239.255.255.250	SSDP	326	NOTIFY * HTTP/1.1
8	2.766974	10.0.0.254	239.255.255.250	SSDP	326	NOTIFY * HTTP/1.1
9	2.767715	10.0.0.254	239.255.255.250	SSDP	381	NOTIFY * HTTP/1.1
10	2.768361	10.0.0.254	239.255.255.250	SSDP	381	NOTIFY * HTTP/1.1
11	2.769103	10.0.0.254	239.255.255.250	SSDP	391	NOTIFY * HTTP/1.1
12	2.769651	10.0.0.254	239.255.255.250	SSDP	391	NOTIFY * HTTP/1.1
15	3.325064	10.0.0.44	75.75.75.75	DNS	77	Standard query 0xc29 A gaia.cs.umass.edu
17	3.348972	75.75.75.75	10.0.0.44	DNS	93	Standard query response 0xc29 A gaia.cs.umass.edu A 128.119.245.12
30	3.427392	10.0.0.44	75.75.75.75	DNS	83	Standard query 0xeda4 A maxcdn.bootstrapcdn.com
31	3.428514	10.0.0.44	75.75.75.75	DNS	79	Standard query 0xa79 A ajax.googleapis.com
35	3.445049	75.75.75.75	10.0.0.44	DNS	135	Standard query response 0xeda4 A maxcdn.bootstrapcdn.com CNAME cds.j3z9t3p6.hwcdn.net A 2...
36	3.448906	75.75.75.75	10.0.0.44	DNS	95	Standard query response 0xa79 A ajax.googleapis.com A 172.217.12.202
521	3.678228	10.0.0.44	75.75.75.75	DNS	75	Standard query 0xdcfa A www.pearson.com
522	3.678393	10.0.0.44	75.75.75.75	DNS	79	Standard query 0xb436 A www.vitalsource.com
523	3.678598	10.0.0.44	75.75.75.75	DNS	72	Standard query 0xd3a3 A redshelf.com
526	3.695928	75.75.75.75	10.0.0.44	DNS	169	Standard query response 0xdcfa A www.pearson.com CNAME wilcard.pearson.com.edgekey.net C...
527	3.698647	10.0.0.44	75.75.75.75	DNS	74	Standard query 0xe1a9 A www.amazon.com
528	3.703716	75.75.75.75	10.0.0.44	DNS	159	Standard query response 0xb436 A www.vitalsource.com A 104.17.67.241 A 104.17.65.241 A 10...
529	3.704968	75.75.75.75	10.0.0.44	DNS	88	Standard query response 0xd3a3 A redshelf.com A 34.196.10.62
530	3.718156	75.75.75.75	10.0.0.44	DNS	169	Standard query response 0xe1a9 A www.amazon.com CNAME tp.47cf2c8c9-frontier.amazon.com CN...
541	6.801907	10.0.0.44	75.75.75.75	DNS	96	Standard query 0x6cf4 A ss-prod-ue1-notif-63.aws.adobess.com
542	6.818616	75.75.75.75	10.0.0.44	DNS	144	Standard query response 0x6cf4 A ss-prod-ue1-notif-63.aws.adobess.com A 52.205.134.231 A ...

Time to Live: 4
Protocol: UDP (17)
Header Checksum: 0xd902 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.0.254
Destination Address: 239.255.255.250

▼ User Datagram Protocol, Src Port: 47931, Dst Port: 1900
Source Port: 47931
Destination Port: 1900
Length: 283
Checksum: 0xf7a8 [unverified]
[Checksum Status: Unverified]
[Stream index: 0]
► [Timestamps]
UDP payload (275 bytes)

▼ Simple Service Discovery Protocol
► NOTIFY * HTTP/1.1\r\n
HOST: 239.255.255.250:1900\r\n

Exercitiul 4:

Stim ca Length este reprezentat pe 2 bytes. Deci valoarea maxima pe care o poate lua acesta este $2^{16} - 1 = 65535$. Astfel, numarul maxim de bytes pe care il poate avea un payload este Length – 8 bytes (lungimea antetului).

Numarul maxim de bytes pe care il poate avea un payload este: 65527.

Exercitiul 5:

Cel mai mare numar de port sursa posibil poate fi calculat asemanator ca la exercitiul anterior. Numarul de port sursa este reprezentat pe 2 bytes, deci cel mai mare numar de port sursa posibil este $2^{16} - 1 = 65535$.

Exercitiul 6:

Pastram filtrul de „udp” si selectam tot primul pachet.

Observam ca numarul de protocol pentru UDP este 17.

udp						
No.	Time	Source	Destination	Protocol	Length	Info
5	2.764896	10.0.0.254	239.255.255.250	SSDP	317	NOTIFY * HTTP/1.1
6	2.765750	10.0.0.254	239.255.255.250	SSDP	317	NOTIFY * HTTP/1.1
7	2.766303	10.0.0.254	239.255.255.250	SSDP	326	NOTIFY * HTTP/1.1
8	2.766974	10.0.0.254	239.255.255.250	SSDP	326	NOTIFY * HTTP/1.1
9	2.767715	10.0.0.254	239.255.255.250	SSDP	381	NOTIFY * HTTP/1.1
10	2.768361	10.0.0.254	239.255.255.250	SSDP	381	NOTIFY * HTTP/1.1
11	2.769103	10.0.0.254	239.255.255.250	SSDP	391	NOTIFY * HTTP/1.1
12	2.769651	10.0.0.254	239.255.255.250	SSDP	391	NOTIFY * HTTP/1.1
15	3.325064	10.0.0.44	75.75.75.75	DNS	77	Standard query 0x3c29 A gaia.cs.umass.edu
17	3.348972	75.75.75.75	10.0.0.44	DNS	93	Standard query response 0x3c29 A gaia.cs.umass.edu A 128.119.245.12
30	3.427392	10.0.0.44	75.75.75.75	DNS	83	Standard query 0xeda4 A maxcdn.bootstrapcdn.com
31	3.428514	10.0.0.44	75.75.75.75	DNS	79	Standard query 0x0a79 A ajax.googleapis.com
35	3.445049	75.75.75.75	10.0.0.44	DNS	135	Standard query response 0xeda4 A maxcdn.bootstrapcdn.com CNAME cds.j3z9t3p6.hwc
36	3.448906	75.75.75.75	10.0.0.44	DNS	95	Standard query response 0x0a79 A ajax.googleapis.com A 172.217.12.202
521	3.678228	10.0.0.44	75.75.75.75	DNS	75	Standard query 0xdcfa A www.pearson.com
522	3.678393	10.0.0.44	75.75.75.75	DNS	79	Standard query 0xb436 A www.vitalsource.com
523	3.678598	10.0.0.44	75.75.75.75	DNS	72	Standard query 0xd3a3 A redshelf.com
526	3.695928	75.75.75.75	10.0.0.44	DNS	169	Standard query response 0xdcfa A www.pearson.com CNAME wilcard.pearson.com.edg
527	3.698647	10.0.0.44	75.75.75.75	DNS	74	Standard query 0xe1a9 A www.amazon.com
528	3.703716	75.75.75.75	10.0.0.44	DNS	159	Standard query response 0xb436 A www.vitalsource.com A 104.17.67.241 A 104.17.6
529	3.704968	75.75.75.75	10.0.0.44	DNS	88	Standard query response 0xd3a3 A redshelf.com A 34.196.10.62

Internet Protocol Version 4, Src: 10.0.0.254, Dst: 239.255.255.250

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 303

Identification: 0xalc3 (41411)

> Flags: 0x40, Don't fragment

0... = Reserved bit: Not set

.1.. = Don't fragment: Set

..0. = More fragments: Not set

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 4

Protocol: UDP (17)

Header Checksum: 0xd902 [validation disabled]

[Header checksum status: Unverified]

Source Address: 10.0.0.254

Destination Address: 239.255.255.250

Exercitiul 7:

Pastram filtrul de „udp” si ne uitam in pachete sa determinam o pereche de pachete in care hostul trimite primul pachet UDP, iar al doilea da reply. Observam ca pachetele cu numerele 15 si 17 indeplinesc copnditiile.

Relatia dintre cele doua pachete este aceea ca hostul are acelasi source port ca destination portul celui de al doilea (58350), iar in mod identic, hostul are acelasi destination port ca source portul celui de al doilea (53).

No.	Time	Source	Destination	Protocol	Length	Info
5	2.764896	10.0.0.254	239.255.255.250	SSDP	317	NOTIFY * HTTP/1.1
6	2.765750	10.0.0.254	239.255.255.250	SSDP	317	NOTIFY * HTTP/1.1
7	2.766303	10.0.0.254	239.255.255.250	SSDP	326	NOTIFY * HTTP/1.1
8	2.766974	10.0.0.254	239.255.255.250	SSDP	326	NOTIFY * HTTP/1.1
9	2.767715	10.0.0.254	239.255.255.250	SSDP	381	NOTIFY * HTTP/1.1
10	2.768361	10.0.0.254	239.255.255.250	SSDP	381	NOTIFY * HTTP/1.1
11	2.769103	10.0.0.254	239.255.255.250	SSDP	391	NOTIFY * HTTP/1.1
12	2.769651	10.0.0.254	239.255.255.250	SSDP	391	NOTIFY * HTTP/1.1
15	3.325064	10.0.0.44	75.75.75.75	DNS	77	Standard query 0x3c29 A gaia.cs.umass.edu
17	3.348972	75.75.75.75	10.0.0.44	DNS	93	Standard query response 0x3c29 A gaia.cs.umass.edu A 128.119.245.12
30	3.427392	10.0.0.44	75.75.75.75	DNS	83	Standard query 0xeda4 A maxcdn.bootstrapcdn.com
31	3.428514	10.0.0.44	75.75.75.75	DNS	79	Standard query 0xa79 A ajax.googleapis.com
35	3.445049	75.75.75.75	10.0.0.44	DNS	135	Standard query response 0xeda4 A maxcdn.bootstrapcdn.com CNAME cds.j3z9t3p6.hwc
36	3.448906	75.75.75.75	10.0.0.44	DNS	95	Standard query response 0xa79 A ajax.googleapis.com A 172.217.12.202
521	3.678228	10.0.0.44	75.75.75.75	DNS	75	Standard query 0xdcfa A www.pearson.com
522	3.678393	10.0.0.44	75.75.75.75	DNS	79	Standard query 0xb436 A www.vitalsource.com
523	3.678598	10.0.0.44	75.75.75.75	DNS	72	Standard query 0xd3a3 A redshelf.com
524	3.680098	75.75.75.75	10.0.0.44	DNS	100	Standard query response 0xb436 A www.vitalsource.com CNAME wildcard.pearson.com edu

> Frame 15: 77 bytes on wire (616 bits), 77 bytes captured (616 bits) on interface en0, id 0
 > Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Maxlinea_80:00:00 (00:50:f1:80:00:00)
 > Internet Protocol Version 4, Src: 10.0.0.44, Dst: 75.75.75.75
 > User Datagram Protocol, Src Port: 58350, Dst Port: 53
 Source Port: 58350
 Destination Port: 53
 Length: 43
 Checksum: 0xc31d [unverified]
 [Checksum Status: Unverified]

No.	Time	Source	Destination	Protocol	Length	Info
5	2.764896	10.0.0.254	239.255.255.250	SSDP	317	NOTIFY * HTTP/1.1
6	2.765750	10.0.0.254	239.255.255.250	SSDP	317	NOTIFY * HTTP/1.1
7	2.766303	10.0.0.254	239.255.255.250	SSDP	326	NOTIFY * HTTP/1.1
8	2.766974	10.0.0.254	239.255.255.250	SSDP	326	NOTIFY * HTTP/1.1
9	2.767715	10.0.0.254	239.255.255.250	SSDP	381	NOTIFY * HTTP/1.1
10	2.768361	10.0.0.254	239.255.255.250	SSDP	381	NOTIFY * HTTP/1.1
11	2.769103	10.0.0.254	239.255.255.250	SSDP	391	NOTIFY * HTTP/1.1
12	2.769651	10.0.0.254	239.255.255.250	SSDP	391	NOTIFY * HTTP/1.1
15	3.325064	10.0.0.44	75.75.75.75	DNS	77	Standard query 0x3c29 A gaia.cs.umass.edu
17	3.348972	75.75.75.75	10.0.0.44	DNS	93	Standard query response 0x3c29 A gaia.cs.umass.edu A 128.119.245.12
30	3.427392	10.0.0.44	75.75.75.75	DNS	83	Standard query 0xeda4 A maxcdn.bootstrapcdn.com
31	3.428514	10.0.0.44	75.75.75.75	DNS	79	Standard query 0x0a79 A ajax.googleapis.com
35	3.445049	75.75.75.75	10.0.0.44	DNS	135	Standard query response 0xeda4 A maxcdn.bootstrapcdn.com CNAME cds.j
36	3.448906	75.75.75.75	10.0.0.44	DNS	95	Standard query response 0x0a79 A ajax.googleapis.com A 172.217.12.20
521	3.678228	10.0.0.44	75.75.75.75	DNS	75	Standard query 0xdcfa A www.pearson.com
522	3.678393	10.0.0.44	75.75.75.75	DNS	79	Standard query 0xb436 A www.vitalsource.com
523	3.678598	10.0.0.44	75.75.75.75	DNS	72	Standard query 0xd3a3 A redshelf.com
526	3.680038	75.75.75.75	10.0.0.44	DNS	160	Standard query response 0xdcfa A www.pearson.com CNAME wildcard.pear

> Frame 17: 93 bytes on wire (744 bits), 93 bytes captured (744 bits) on interface en0, id 0
 > Ethernet II, Src: Maxlinea_80:00:00 (00:50:f1:80:00:00), Dst: Apple_98:d9:27 (78:4f:43:98:d9:27)
 > Internet Protocol Version 4, Src: 75.75.75.75, Dst: 10.0.0.44
 > User Datagram Protocol, Src Port: 53, Dst Port: 58350

Source Port: 53
 Destination Port: 58350
 Length: 90
 Checksum: 0x4af2 [unverified]
 [Checksum Status: Unverified]
 [Stream index: 1]